WHITEPAPER: Alepic
(Alephium Collaborative Canvas)

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Platform: Alephium Blockchain
Format: Public Digital Asset & Interactive Art Protocol
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1. Executive Summary

The **Alepic** (Alephium Collaborative Canvas) is a decentralized, community-driven digital art platform designed to create and manage a single, massive-scale public image hosted on blockchain technology. Unlike static NFT projects or centralized social media feeds, **Alepic** allows thousands of users to collectively own and edit sections of a unified 4K artwork in real-time.

Key Features:

- **Resolution:** Fixed 4096×2160 pixels (UHD/4K) suitable for global billboard deployment.
- **Ownership Model:** The image is divided into **34,560 unique chunks** (16×16 pixels each). Each chunk has a single owner who controls edits within their territory.
- **Dynamic Pricing:** Initial unclaimed chunks follow a fair ramp-up pricing model starting at 1 ALPH, increasing by 1 ALPH for each subsequent purchase to prevent speculation and ensure equitable access.
- **On-Chain History:** Every color change and ownership transfer is recorded immutably on the Alephium blockchain.
- **Ecosystem Incentives:** A robust partner program rewards third-party developers (web sites/apps) for driving traffic to the platform via a revenue-sharing model from transaction fees.
- **Engagement:** Integrated **"Alepe"** game mechanic rewards active participants and sustains the ecosystem's economy through a **Treasury** system.

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2. Problem Statement

2.1 The Static Art Limitation

Traditional digital art platforms (e.g., standard NFT marketplaces) suffer from rigidity. Once an asset is minted, its visual content rarely changes without creating a new token, leading to fragmented history and lost context for collaborative works.

2.2 Centralized Censorship Risks

Most public displays rely on centralized servers. If the server goes offline or is censored, the artwork disappears. This poses risks for long-term projects like global billboard campaigns that require resilience and uptime.

2.3 Lack of Community Ownership

Current platforms rarely allow true shared ownership of a single visual narrative. Users are passive viewers rather than active co-creators with verifiable rights to specific areas of the canvas.

2.4 Short-Lived Creations

Alepic provides eternal storage for the social art experiment which will exponentially gain its historical value over decades.

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3. Solution: The Collaborative Canvas Protocol

Alepic transforms the digital artwork into a dynamic, living entity where ownership is fragmented but unified under a single protocol on the Alephium blockchain.

3.1 Image Specification

To ensure high visual fidelity for billboard displays while maintaining blockchain efficiency:

- **Fixed Resolution:** 4096×2160 pixels (8,847,360 total pixels).
- **Color Depth:** Optimized 4-bit Indexed Palette to balance quality and on-chain storage costs.
- **Global Consistency:** The image is rendered identically for all users based on the consensus of ownership records.

3.2 Chunk-Based Ownership Architecture

The canvas is logically divided into a grid of small territories called **Chunks**.

Parameter	Value	Description
Chunk Size	16×16 pixels	256 pixels per chunk
Total Chunks	34,560	Grid of 256×135 chunks
Ownership Rights	Singular	Each chunk has exactly one owner address at any time.
Edit Privilege	Exclusive	Only the owner can modify pixel colors within their assigned chunk.

3.3 Decentralized History & Reconstruction

To overcome blockchain storage limits, the image state is not stored as a single file but reconstructed from a history of transactional blocks.

- **Storage:** Each block (max 32 KB on Alephium) stores updates for multiple chunks or their metadata hashes.
- **Reconstruction:** Clients download the latest history and map chunk states to reconstruct the image in real-time.
- **Integrity:** A Merkle root of the global state is periodically anchored, allowing anyone to verify the integrity of the display without trusting a central operator.

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4. The Mascot: Alepe Game Layer

To incentivize participation and maintain community engagement, a randomized game mechanic featuring the official mascot **"Alepe"** is embedded in the contract:

- **Starting Position:** Each new session begins with **Alepe** located at the **central pixel** of the canvas (x=2048; y=1080).
- **Trigger Interval:** Every **100,000 blocks**, **Alepe** jumps to a random location within the canvas boundaries. At ~1 block/second on Alephium, this corresponds to approximately **27 hours 46 minutes** (roughly once every day and a half).
- **Wrap-Around Logic:** If the jump exceeds canvas edges, it teleports to the opposite side using modulo arithmetic:
  ```
  x = (x + distance*cos(angle)) % WIDTH
  y = (y + distance*sin(angle)) % HEIGHT
  ```
- **Reward Condition:** The owner of the chunk where **Alepe** lands receives a share of the **Treasury** (1% of total balance). **If Alepe lands on an unclaimed chunk, no reward is issued and funds remain in the Treasury.**

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5. Technical Architecture

5.1 Blockchain Backend: Alephium

The project utilizes the **Alephium** blockchain due to its high throughput (~1 block/second) and smart contract capabilities (AluX).

- **Transaction Limit:** 32 KB per block is strictly enforced. Updates are batched or fragmented across consecutive blocks if a chunk update exceeds single-block capacity.
- **State Storage:** The smart contract maintains:
  – Price for the next unclaimed chunk;
  – Chunk ID to Owner Address mapping;
  – **Alepe** game state and rewards.
- **Transaction Fees & Referral Tracking:** Every transaction includes an optional referrer address field. If provided, a portion of the fee is automatically routed to this address via internal transfer logic within the contract.

5.2 Client-Side Rendering (React Native)

The front-end is built using **React Native** for cross-platform compatibility (Web, Android, iOS).

- **Dynamic Loading:** Users only download chunks currently visible on their viewport to optimize bandwidth.
- **Visualization:** A custom rendering engine displays the 4K canvas with a zoomable interface.
- **Live Events:** WebSocket-like polling listens for blockchain events (e.g., "Chunk Updated" or "**Alepe** Jumped") and triggers real-time UI animations.

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6. Economic Model & Tokenomics

6.1 Dynamic Pricing for Initial Ownership

To ensure fair access during the initial distribution phase, unclaimed chunks follow a progressive pricing model:

Metric	Value	Description
Initial Price	1 ALPH	First user to claim any unclaimed chunk pays this base price.
Increment per Claim	+1 ALPH	Each subsequent unclaimed chunk purchase increases by exactly 1 ALPH.
Price Cap	~34,560 ALPH	Theoretical maximum price for the last unclaimed chunk (fair distribution).

This mechanism prevents speculative hoarding and ensures early adopters receive equitable access before prices scale with demand.

6.2 Treasury System & Fee Distribution

To ensure sustainability, community growth, and developer incentives, a tiered fee distribution model is implemented:

Recipient	Share of Initial Sale Proceeds	Share of Secondary Transaction Fee	Purpose
Platform Treasury	100% (all sale proceeds)	4.0% (80% of Transaction Fee)	**Alepe** rewards
Partner/Referrer Wallet	20% of the 5% Transaction Fee	1.0% (20% of Transaction Fee)	Incentivizes third-party apps & clients

**Transaction Fee is paid by buyer*

- **Initial Sale (Unclaimed Chunks):** All sale proceeds from first-time chunk purchases are routed directly to the **Treasury**. No individual seller receives funds for initial claims.
- **Secondary Sales:** A **5% tax** is levied on every secondary sale or ownership transfer upon execution. The smart contract automatically splits the collected fee:
  – If a valid referrer address is attached to the transaction (e.g., via a specific client app), **20%** of the total fee (1% of value) is transferred from the **Treasury** immediately upon confirmation.
  – The remaining **80%** of the fee (4% of value) stays in the main **Treasury** to fund operations and rewards.

6.3 Alepe Reward Logic for Unclaimed Chunks

To prevent free-riding on unowned territory:

- If **Alepe** lands on a chunk that has no registered owner, **no reward is issued**.
- The funds intended for the reward remain in the **Treasury** to support future rewards for active claimants.

6.4 Developer Ecosystem Incentives

Alepic actively encourages the creation of third-party tools, web interfaces, and mobile clients that provide alternative access points to the network.

- **Referral Wallets:** Developer can provide an optional referrer wallet address field in the transaction to get reward.
- **Passive Income:** Users interacting with these tools automatically route the fee portion to the developer's wallet, creating a sustainable revenue stream for community builders.

6.5 Additional Incentives & Governance

- Chunk owners are automatically motivated to show the Canvas on a big city street – that will exponentially raise the price of a Chunk.
- **No Governance.**

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7. Security & Privacy

7.1 Data Integrity

- **Immutable History:** All changes are cryptographically signed and stored in blocks. No central administrator can alter the history or "erase" edits.
- **Signature Verification:** Every edit requires a valid digital signature from the chunk owner, preventing unauthorized modifications.
- **Fee Split Security:** The referral fee logic is hardcoded into the consensus layer of the contract, ensuring that payments to partners cannot be manipulated by any single node operator.

7.2 Privacy

- Privacy is provided by Alephium network.

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8. Scalability & Global Deployment

8.1 Billboard Integration

The architecture is designed for offline digital signage:

- **QR Activation:** Physical billboards use unique QR codes to connect local display units securely to the network node.
- **Low-Latency Sync:** Billboards periodically sync with the blockchain to fetch the latest chunk states, ensuring the displayed image matches the consensus.

8.2 Future Upgrades

Only client code will be upgraded.

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9. Roadmap

Phase	Milestones	Timeline
Phase 1	Smart contract development, Chunk ownership logic implementation	Q2 2026
Phase 2	Test smart contract deployment	Q3 2026
Phase 3	React Native web-client development & official client launch (+GitHub-release)	Q4 2026
Phase 4	Official release	Q1 2027
Phase 5	Android application development & launch - TBD (+GitHub-release)	Q2 2027
Phase 6	IOS application development & launch - TBD (+GitHub-release)	Q3 2027

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10. Conclusion

The **Alepic** represents a paradigm shift in digital ownership and public art. By combining the permanence of blockchain with the dynamism of real-time collaboration, **Alepic** empowers communities to create art that is truly theirs—immutable yet evolving. With robust technical foundations on Alephium, a sustainable economic model rewarding both creators and developers, and a vision for global billboard integration featuring the mascot **Alepe**, **Alepic** is poised to become the standard for decentralized visual experiences.

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